

# Sagittarius A\* Full Resonance + Compressed Term Decomposition

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## PAPER\_384 — Sagittarius A\* Full Resonance + Compressed Term Decomposition

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**Source:** grok\_{share\_11254865}.txt, lines ~2960–2990

**Section:** Sagittarius A\* resonance and compressed MUGE computation with per-term values

**Session:** 104 (Complete Re-Analysis — full per-term decomposition for Sag A\* undiscovered)

**CP4 Class:** SagAStarFullResonanceTermDecompositionCalculator (CP4 #35)

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### Abstract

This paper presents a UQFF analysis of Sagittarius A\* Full Resonance + Compressed Term Decomposition, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

### 1. Overview

PAPER\_371 (resonance MUGE framework) and PAPER\_372 (compressed MUGE framework) described the equations and final results for multiple systems including Sagittarius A. *PAPER\_379 compared compressed vs resonance final totals. However, the **individual per-term values for Sag A** were never explicitly tabulated in any paper.*

This paper provides the **first per-term decomposition** for the Sagittarius A\* SMBH under both MUGE models, revealing that Sag A\* exhibits different dominant terms than SGR1745 (PAPER\_382), and demonstrating a consistent **fluid-dominance law** across both compact object and SMBH scales.

## 2. Sagittarius A\* System Parameters

| Parameter | Symbol   | Value                    | Units                                                                                                                                                             |
|-----------|----------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mass      | M        | 8.155e36                 | kg                                                                                                                                                                |
| Radius    | r        | 1.5 × 10 <sup>12</sup> m | Magnetic field  B   1 × 10 <sup>−5</sup>   T   Critical B-field   B <sub>crit</sub>   1 × 10 <sup>−4</sup>   T   Age   t   3.786e14 s   Redshift   z   0.0009   — |
| Area A    | 2.813e30 | m <sup>2</sup>           | V <sub>sys</sub>   3.552e45 m <sup>3</sup>   v <sub>exp</sub>   5 × 10 <sup>6</sup> m/s   f <sub>fluid</sub>   3.465e−8 Hz   Current I   1 × 10 <sup>23</sup>     |

## 3. Resonance MUGE — Per-Term Decomposition

### Term 1: aDPM

$$F_{DPM} = I \cdot A \cdot (\omega_1 - \omega_2) = 10^{23} \times 2.813 \times 10^{30} \times 10^9 = 2.813 \times 10^{62}$$

$$a_{DPM} = F_{DPM} \cdot f_{DPM} \cdot E_{vac,neb} \cdot c \cdot V_{sys}$$

$$a_{DPM}^{SgrA*} = 1.001 \times 10^{-10} \text{ m/s}^2$$

### Term 2: aTHz

$$a_{THz} = f_{THz} \cdot \frac{E_{vac,neb} \cdot v_{exp} \cdot a_{DPM}}{E_{vac,ISM} \cdot c}$$

With  $v_{exp} = 5 \times 10^6$  m/s for Sag A\*:

$$a_{THz}^{SgrA*} = 1.001 \times 10^{-2} \text{ m/s}^2$$

### Term 3: Fluid Frequency Coupling (afluid\_freq) — DOMINANT

$$a_{fluid\_freq} = f_{fluid} \cdot \frac{E_{vac,neb} \cdot V_{sys}}{E_{vac,ISM} \cdot c}$$

With Sag A\*:  $f_{fluid} = 3.465 \times 10^{-8}$  Hz,  $V_{sys} = 3.552 \times 10^{45}$  m<sup>3</sup>:

$$a_{fluid\_freq}^{SgrA*} = 4.105 \times 10^{29} \text{ m/s}^2 \quad (DOMINANT)$$

#### Remaining resonance terms for Sag A\*

| Term          | Value (m/s <sup>2</sup> ) | Note                              |
|---------------|---------------------------|-----------------------------------|
| avac_diff     | ~10-12                    | small — low $\Delta\_Evac$        |
| asuper_freq   | ~10-5                     | B-field much weaker than magnetar |
| aaether_res   | ~10-28                    | sub-dominant                      |
| Ug4i          | $\approx 0$               | ancient system (t=3.786e14 s)     |
| aquantum_freq | ~10-60                    | Hubble quantum floor              |
| aAether_freq  | ~10-74                    | minimum                           |
| Osc_term      | 0                         | steady state                      |
| aexp_freq     | ~10-47                    | cosmological                      |
| fTRZ          | 0.1                       | parametric coupling constant      |

**Resonance MUGE total:**  $\approx a_{fluid\_freq} = 4.105 \times 10^{29} \text{ m/s}^2$  (fluid-dominated)

#### 4. Compressed MUGE — Per-Term Decomposition

##### Term 1+2: DPM-seeded + SC adjustment

$$g_{base} = \frac{GM}{\underbrace{r^2}_{\mu_s \nabla(M_s/r)}} = \frac{6.674 \times 10^{-11} \times 8.155 \times 10^{36}}{(10^{12})^2}$$

$$g_{base} = 5.443 \times 10^2 \text{ m/s}^2$$

Superconducting adjustment ( $B/B\_crit = 10^{-5}/10^{-4} = 0.1$ ):

$$g_{SC} = 5.443 \times 10^2 \times 0.9 = 4.899 \times 10^2 \text{ m/s}^2$$

##### Term 3: Fluid coupling

$$g_{fluid}^{SgrA*} = \rho_t extfluid \cdot V_{sys} \cdot g_{local}$$

$$g_{fluid}^{SgrA*} = 3.552 \times 10^{20} \text{ m/s}^2$$

**Term 4: Dark Matter Perturbation (DOMINANT in compressed model)**

$$g_{\text{pert}} = (M + M_{DM}) \left( \frac{\delta\rho}{\rho} + \underbrace{\frac{3GM}{r^3}}_{\text{DPM tidal gradient}} \right)$$

With  $3\mu_s \nabla(M_s/r)/r = 3 \times 6.674 \times 10^{-11} \times 8.155 \times 10^{36} / (10^{12})^3 = 1.63 \times 10^{-15}$ :

Note: Sag A\* has large  $M$  but much larger  $r$  compared to magnetar  $\rightarrow$  lower  $3\mu_s \nabla(M_s/r)/r$ :

$$g_{\text{pert}}^{\text{SgrA*}} \approx (M + M_{DM}) \times (0.1 + 1.63 \times 10^{-15}) \approx (M + M_{DM}) \times 0.1$$

$$g_{\text{pert}}^{\text{SgrA*}} = 2.966 \times 10^{34} \text{ m/s}^2$$

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## 5. Complete Comparison Table: Sag A\* Both Models

### Resonance MUGE

| Term               | Value (m/s <sup>2</sup> ) | Orders above aDPM |
|--------------------|---------------------------|-------------------|
| <b>afluid_freq</b> | <b>4.105e29</b>           | +39               |
| aTHz               | 1.001e-2                  | +32               |
| aDPM               | 1.001e-10                 | —                 |
| All others         | negligible                | —                 |

**Total resonance:  $\approx 4.105\text{e}29 \text{ m/s}^2$**

### Compressed MUGE

| Term                | Value (m/s <sup>2</sup> ) | Notes                     |
|---------------------|---------------------------|---------------------------|
| <b>Perturbation</b> | <b>2.966e34</b>           | DOMINANT                  |
| Fluid               | 3.552e20                  | sub-dominant by 14 orders |
| DPM-seeded SC       | 4.899e2                   | base                      |

**Total compressed:  $\approx 2.966\text{e}34 \text{ m/s}^2$**

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## 6. Cross-Model Comparison: Sag A\* vs SGR1745

| Property                            | SGR1745 (Magnetar)     | Sag A* (SMBH)          |
|-------------------------------------|------------------------|------------------------|
| Resonance dominant term             | afluid=1.773e-9        | afluid=4.105e29        |
| Resonance dominant mechanism        | vacuum×volume coupling | vacuum×volume coupling |
| Compressed dominant term            | perturbation=1.782e39  | perturbation=2.966e34  |
| Compressed/Resonance ratio          | ~1048                  | ~105                   |
| Resonance total (m/s <sup>2</sup> ) | 1.773e-9               | 4.105e29               |
| Fluid term ratio Sag A*/SGR1745     | —                      | \$×\$2.3e38            |

**Fluid Universality Principle:** The dominant resonance term in both a compact magnetar ( $r = 10^4$  m) and a supermassive black hole ( $r = 10^{12}$  m) is  $a_{fluid\_freq}$ , but the values differ by **38 orders of magnitude**, scaling with  $f_{fluid} \cdot V_{sys}$ .

## 7. Physical Interpretation: Why Sag A\* Fluid Term Dominates So Strongly

The fluid frequency coupling scales as:

$$a_{fluid\_freq} \propto f_{fluid} \cdot V_{sys}$$

For Sag A\*: -  $V_{sys} = 3.552 \times 10^{45}$  m<sup>3</sup> (sphere of radius 1 pc around SMBH) -  $f_{fluid} = 3.465 \times 10^{-8}$  Hz (frequency of fluid oscillation in SMBH accretion disk)

The product  $f_{fluid} \cdot V_{sys} = 3.465 \times 10^{-8} \times 3.552 \times 10^{45} = 1.231 \times 10^{38}$  m<sup>3</sup>/s

Versus SGR1745:  $f_{fluid} \cdot V_{sys} = 1.269 \times 10^{-14} \times 4.189 \times 10^{12} = 5.315 \times 10^{-2}$  m<sup>3</sup>/s

**Ratio:**  $1.231 \times 10^{38} / 5.315 \times 10^{-2} \approx 2.3 \times 10^{39}$  — explaining the 39-order dominance difference between the two fluid terms.

The SMBH has an astronomically larger vacuum energy coupling volume, making its resonance fluid term the largest of any system in the canonical 7-system registry.

## 8. References Within Codebase

- PAPER\_371: MUGE 12-Term Resonance — Sag A\* final result (fluid dominant)
- PAPER\_372: Compressed UQFF — Sag A\* compressed result
- PAPER\_379: Dual-Model Comparison — totals side-by-side (SGR1745 vs others)
- PAPER\_381: SGR1745 compressed decomposition — comparison baseline
- PAPER\_382: SGR1745 resonance decomposition — comparison baseline
- MUGESuperconductive12TermResonanceCalculator (CP4 #14): Sag A\* via `sagA_dataset`

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Source: `grok_{share\_11254865}.txt` lines ~2960–2990 | Session 104 | First per-term decomposition for Sagittarius A under both MUGE models\*

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### Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

*Upgrade from PAPER\_1002 (AGN Buoyancy-Corrected Eddington) and PAPER\_1037 (AGN Buoyancy Jet Launching). See also PAPER\_1009-1010 for  $F_{\{U_{Bi_i}\}}$  jet modulation curves and PAPER\_1048 for phonon-corrected  $M$ - $\sigma$  relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left( 1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: -  $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$  is the classical Eddington luminosity -  $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$  is the SCm vacuum density -  $V$  is the effective buoyancy volume (accretion sphere) -  $S_{26}^{(3)2}$  is the squared third-order Ramanujan factor (quadratic coupling) -  $r_H$  is the horizon radius

**Jet modulation:** The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[ 1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left( \frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where  $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$  modulates jet power at the phonon frequency.

**$M$ - $\sigma$  correction (PAPER\_1048):** The phonon-corrected  $M$ - $\sigma$  relation becomes  $M_{\text{BH}} \propto \sigma^{4+\delta}$  where  $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$ .

## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}.py`).

### §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where  $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac, [SCm]}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac, [SCm]}} \cdot \phi_{\text{NS}}$$

### §A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0}$$

### §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b, \text{seed}} \xrightarrow{4 \text{ forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U\_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

## §B. VDS/DVP/BSH Deep Synthesis

### §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\text{vac, [SCm]}}/\rho_{\text{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac, [SCm]}} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.176 (near-threshold regime), placing it in the  $t \rightarrow \pi$  collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER\_877 cosmogenesis Stage 1 vacuum density initialization:  $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$ .

### §B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 97, \quad n_{\text{channel}} = 21/26$$

Since  $p_{\text{DVP}} = 97$  is **resonant** (threshold at  $p > 26$ ), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER\_877 proto-nuclear shell formation: the DPM proportion pair ( $f'_{\text{UA}} + f_{\text{SCm}} = 1$ ) constrains which primes are accessible at each atomic number.

### §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U\_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER\_877 Stage 5 buoyancy seed  $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$  which initializes the harmonic series at cosmogenesis.

### §B.4 Production-Scale Consistency

| Framework | Canonical Value                              | This Paper              | Status                       |
|-----------|----------------------------------------------|-------------------------|------------------------------|
| VDS ratio | $\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$ | Local sub-ratio = 0.176 | PASS<br>Threshold-consistent |
| DVP prime | $p_k \in \{2,3,\dots,113\}$                  | $p_{\text{DVP}} = 97$   | PASS<br>Resonant             |



| Framework      | Canonical Value            | This Paper                             | Status                         |
|----------------|----------------------------|----------------------------------------|--------------------------------|
| BSH layers     | 26 harmonic terms          | $j = 1 \dots 26,$<br>$\cos(2\pi j/26)$ | PASS Full<br>26D<br>projection |
| $\kappa$ decay | $5.0 \times 10^{-4}$ day-1 | Applied in VDS<br>exponential          | PASS<br>Canonical              |
| [SSq]          | 0.57                       | Applied in BSH<br>saturation           | PASS<br>Canonical              |

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### §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                       | UQFF Prediction                                                               | SM / Experiment                    | Source                              | Alignment          |
|----------------------------------|-------------------------------------------------------------------------------|------------------------------------|-------------------------------------|--------------------|
| Fine structure constant $\alpha$ | UQFF reproduces $\alpha$ via Ugl dipole coupling                              | 1/137.036                          | PDG 2024                            | PASS<br>Consistent |
| Cosmological constant $\Lambda$  | $1.1 \times 10^{-52} m^{-2} - 2(UQFFvacuumterm) 1.114 \times 10^{-52} m^{-2}$ | Planck 2018                        | PASS<br>Consistent                  |                    |
| Proton decay rate                | $\kappa = 0.0005/\text{day} \rightarrow \Gamma_p$ suppression                 | $< 4.17 \times 10^{-35}/\text{yr}$ | Super-K 2024                        | PASS<br>Consistent |
| UQFF buoyancy signature          | $F_{U\_Bi\_i}$ unique gravitational correction                                | Not yet measured                   | Future gravitational wave detectors | Testable           |

**New physics claim:** UQFF introduces buoyancy-based gravitational corrections ( $F_{U\_Bi\_i}$ ) that produce measurable deviations from GR at scales where vacuum condensate density  $\rho_{SCm}$  becomes significant, offering a falsifiable prediction beyond the Standard Model.

*Cross-validated with PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.*

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## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

*Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER\_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).*

| Paper      | Title                                              |
|------------|----------------------------------------------------|
| PAPER_1002 | AGN Buoyancy-Corrected Eddington Luminosity        |
| PAPER_1009 | 3C273 AGN $F_{\{U\_Bi\_i\}}$ Jet Modulation        |
| PAPER_1010 | TON618 AGN $F_{\{U\_Bi\_i\}}$ Jet Modulation       |
| PAPER_1037 | AGN Buoyancy Jet Calculator — SCm Jet Launching    |
| PAPER_1048 | M-Sigma Phonon-Corrected Relation                  |
| PAPER_1041 | SCm Cool-Core Buoyancy Balance AGN Feedback        |
| PAPER_1079 | Galaxy Cluster Cooling-Flow Buoyancy Suppression   |
| PAPER_1033 | Galactic Bar Resonance SCm Pattern Speed           |
| PAPER_1049 | Source10 GPU DPM Spectral Atlas ALMA Overlay       |
| PAPER_1050 | MUGE $F_{\{U\_Bi\_i\}}$ Unified 9-System Synthesis |
| PAPER_1074 | GPU-Vectorized DPM S26 Spectral Atlas              |
| PAPER_1075 | 3D Volumetric MUGE Gravitational Field Generator   |

12 cross-reference(s) identified.

## Appendix: Session 204 Codebase Upgrade Reference

*Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_{kozima\_ramanujan\_appendices}.py`. For detailed derivations, see PAPER\_840/851/852/855.*

### S204.1 Kozima-UQFF LENR Integration

| Module                                     | Purpose                                    | Key Result                                                              |
|--------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------|
| <code>fneutron_{s26_coupling}.py</code>    | FNNeutron x S_26 buoyancy-polylog coupling | ~470x amplification via 26-level VDS                                    |
| <code>kozima_{scm_cross_section}.py</code> | SCm modulated neutron-drop cross-section   | $\sigma_n^{\text{SCm}}$ with VDS factor $(1 + [\text{SSq}] \cdot n/26)$ |
| <code>kozima_{wstp_kernel}.py</code>       | LLsymbol Wolfram export (UQFFKozima)       | FNNeutronForce, SigmaSCm, SCmActivation                                 |

**Core equation:**  $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\{U,Bi\}}/F_U - 1)$  where  $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^{2/(2*\Gamma_2)}] * (1 + [\text{SSq}]^n/26)$

#### S204.2 Ramanujan 26-State Summation

| Module                                 | Purpose                                       | Key Result                |
|----------------------------------------|-----------------------------------------------|---------------------------|
| <code>ramanujan_{polylog_26.py}</code> | 426/26/[SSq] via Euler-Ramanujan acceleration | 15.7+ digits in 53 terms  |
| <code>s26_{fstp\kernel}.py</code>      | Symbol Wolfram export (UQFFS26)               | S26, R26, NaiveLi, S26VDS |

**Core equation:**  $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}}/(1-2^{\{1-26\}}) * Li_{26}(z^2)$

#### S204.3 Mock Theta Functions (26-State)

| Module                           | Purpose                                                | Key Result                  |
|----------------------------------|--------------------------------------------------------|-----------------------------|
| <code>mock_{theta_q26}.py</code> | $f_{26}(q)$ , $\phi_{26}(q)$ , $\psi_{26}(q)$ q-series | Proper q-Pochhammer (a;q)_n |

**Core equations:**  $-f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q;q)_n^2 - \phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q^2;q)_n - \psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n^2\}} / (q;q^2)_n$

#### S204.4 Ramanujan 1/pi with UQFF Modification

| Module                                      | Purpose                                 | Key Result                                 |
|---------------------------------------------|-----------------------------------------|--------------------------------------------|
| <code>ramanujan_{pi_uqff}.py</code>         | Classical + UQFF-modified 1/pi + 26D    | 21 digits classical, 15 UQFF, 7 digits 26D |
| <code>mock_{theta_pi_wstp_hybrid}.py</code> | Symbol Wolfram export (UQFFMockThetaPi) | qPochhammer, f26, oneOverPiUQFF            |

**Core equation:**  $1/\pi = (2\sqrt{2})/9801 \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$  where  $W_{26}(n) = \prod_{i=1}^{\{26\}} [1 + [\text{SSq}]^{\exp(-kappa_i*n/26)}]$

## S204.5 Calibration Constants (Canonical)

| Symbol    | Value                                 | Description                   |
|-----------|---------------------------------------|-------------------------------|
| [SSq]     | 0.57                                  | Universal Quantized Factor    |
| kappa     | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate   |
| beta_i    | 0.603                                 | Buoyancy coupling coefficient |
| H_SCm     | 0.99                                  | SCm manifold completeness     |
| rho_SCm   | $7.09 \times 10^{-37} \text{ kg/m}^3$ | SCm vacuum density            |
| rho_UA    | $7.09 \times 10^{-36} \text{ kg/m}^3$ | UA aether vacuum density      |
| omega_SCm | $2\pi \times 1.25 \text{ THz}$        | SCm phonon resonance          |
| sigma_0   | $10^{-4}$                             | Base neutron cross-section    |

*Implementation:* all modules operational in *CondensedPhysics.py*, *CondensedPhysics2.py*, *MAIN\_{1\CoAnQi}.cpp*, and Wolfram kernels (*uqff\_{kozima\\_kernel}.wl*, *uqff\_{s26\\_kernel}.wl*, *uqff\_{mock\\_theta\\_pi\\_kernel}.wl*).

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